

DATABASE TECHNOLOGIES ASSIGNMENT 6

1. Use Group by clause with appropriate SQL functions to solve following queries. Note: To solve below queries, use “sales” database 1. Write a query that counts the number of salespeople registering orders for each day. (If a salesperson has more than one order on a given day, he or she should be counted only once.).

```
Windows PowerShell x Windows PowerShell x Windows PowerShell x + v
5 rows in set (0.029 sec)
KD4_98828_Gayatri@localhost>SELECT odate, COUNT(DISTINCT snum) AS NUMBER_OF_SALESPERSONS
-> FROM orders
-> GROUP BY odate;
+-----+-----+
| odate | NUMBER_OF_SALESPERSONS |
+-----+-----+
| 1990-10-03 | 4 |
| 1990-10-04 | 3 |
+-----+-----+
2 rows in set (0.020 sec)
KD4_98828_Gayatri@localhost>SELECT CONCAT('For the city ', city,
-> ' ', the highest rating is: ', MAX(rating))
-> AS RESULT
-> FROM customers
-> GROUP BY city;
+-----+-----+
| RESULT |
+-----+-----+
| For the city London, the highest rating is: 100 |
| For the city Rome, the highest rating is: 200 |
| For the city San Jose, the highest rating is: 300 |
| For the city Berlin, the highest rating is: 300 |
+-----+-----+
4 rows in set (0.018 sec)
KD4_98828_Gayatri@localhost>SELECT odate, SUM(amount) AS TOTAL_AMOUNT
-> FROM orders
-> GROUP BY odate
-> ORDER BY TOTAL_AMOUNT DESC;
ERROR 1054 (42S22): Unknown column 'amount' in 'field list'
KD4_98828_Gayatri@localhost>SELECT odate, SUM(amount) AS TOTAL_AMOUNT
-> FROM orders
-> GROUP BY odate
-> ^CLECT odate, SUM(amount) AS TOTAL_AMOUNT
```

2. Write a query on the Customers table that will find the highest rating in each city. Put the output in this form: For the city (city), the highest rating is: (rating)

```
Windows PowerShell x Windows PowerShell x Windows PowerShell x + v
-> GROUP BY snum
-> HAVING SUM(amount) > 2565.84;
ERROR 1054 (42S22): Unknown column 'amount' in 'field list'
KD4_98828_Gayatri@localhost>SELECT ODATE, SUM(AMT) AS TOTAL_ORDER_AMOUNT
-> FROM ORDERS
-> GROUP BY ODATE
-> ORDER BY TOTAL_ORDER_AMOUNT DESC;
+-----+-----+
| ODATE | TOTAL_ORDER_AMOUNT |
+-----+-----+
| 1990-10-04 | 16713.81 |
| 1990-10-03 | 8944.59 |
+-----+-----+
2 rows in set (0.009 sec)

KD4_98828_Gayatri@localhost>SELECT SNUM, SUM(AMT) AS TOTAL_AMOUNT
-> FROM ORDERS
-> GROUP BY SNUM
-> HAVING SUM(AMT) > 2565.84;
+-----+-----+
| SNUM | TOTAL_AMOUNT |
+-----+-----+
| 1001 | 15382.07 |
| 1002 | 5546.15 |
+-----+-----+
2 rows in set (0.009 sec)

KD4_98828_Gayatri@localhost>HAVING SUM(AMT) > 2565.84
-> ^C
KD4_98828_Gayatri@localhost>SELECT CITY, MAX(RATING) AS HIGHEST_RATING
-> FROM CUSTOMERS
-> GROUP BY CITY;
+-----+-----+
| CITY | HIGHEST_RATING |
+-----+-----+
| London | 100 |
| Rome | 200 |
+-----+-----+
```

3 Write an SQL query to calculate the total order amount for each day and display the results in descending order of total orders.

```
Windows PowerShell x Windows PowerShell x Windows PowerShell x + v
-> GROUP BY snum
-> HAVING SUM(amount) > 2565.84;
ERROR 1054 (42S22): Unknown column 'amount' in 'field list'
KD4_98828_Gayatri@localhost>SELECT ODATE, SUM(AMT) AS TOTAL_ORDER_AMOUNT
-> FROM ORDERS
-> GROUP BY ODATE
-> ORDER BY TOTAL_ORDER_AMOUNT DESC;
+-----+-----+
| ODATE | TOTAL_ORDER_AMOUNT |
+-----+-----+
| 1990-10-04 | 16713.81 |
| 1990-10-03 | 8944.59 |
+-----+-----+
2 rows in set (0.009 sec)

KD4_98828_Gayatri@localhost>SELECT SNUM, SUM(AMT) AS TOTAL_AMOUNT
-> FROM ORDERS
-> GROUP BY SNUM
-> HAVING SUM(AMT) > 2565.84;
+-----+-----+
| SNUM | TOTAL_AMOUNT |
+-----+-----+
| 1001 | 15382.07 |
| 1002 | 5546.15 |
+-----+-----+
2 rows in set (0.009 sec)

KD4_98828_Gayatri@localhost>HAVING SUM(AMT) > 2565.84
-> ^C
KD4_98828_Gayatri@localhost>SELECT CITY, MAX(RATING) AS HIGHEST_RATING
-> FROM CUSTOMERS
-> GROUP BY CITY;
+-----+-----+
| CITY | HIGHEST_RATING |
+-----+-----+
| London | 100 |
| Rome | 200 |
+-----+-----+
```

4. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the average amount of the order in the table. (Note Use the average amount value directly →2565.84)

```
Windows PowerShell x Windows PowerShell x Windows PowerShell x + v

KD4_98828_Gayatri@localhost>SELECT SNUM, SUM(AMT) AS TOTAL_AMOUNT
-> FROM ORDERS
-> GROUP BY SNUM
-> HAVING SUM(AMT) > 2565.84;
+-----+-----+
| SNUM | TOTAL_AMOUNT |
+-----+-----+
| 1001 | 15382.07 |
| 1002 | 5546.15 |
+-----+-----+
2 rows in set (0.009 sec)

KD4_98828_Gayatri@localhost>HAVING SUM(AMT) > 2565.84
-> ^C
KD4_98828_Gayatri@localhost>SELECT CITY, MAX(RATING) AS HIGHEST_RATING
-> FROM CUSTOMERS
-> GROUP BY CITY;
+-----+-----+
| CITY | HIGHEST_RATING |
+-----+-----+
| London | 100 |
| Rome | 200 |
| San Jose | 300 |
| Berlin | 300 |
+-----+-----+
4 rows in set (0.011 sec)

KD4_98828_Gayatri@localhost>SELECT SNUM, MAX(AMT) AS MAX_ORDER_AMOUNT
-> FROM ORDERS
-> WHERE AMT > 3000
-> GROUP BY SNUM;
+-----+-----+
| SNUM | MAX_ORDER_AMOUNT |
+-----+-----+
| 1002 | 5160.45 |
| 1001 | 9891.88 |
+-----+-----+
```

5. Write a query that selects the highest rating in each city.

```
Windows PowerShell x Windows PowerShell x Windows PowerShell x + v

KD4_98828_Gayatri@localhost>SELECT SNUM, SUM(AMT) AS TOTAL_AMOUNT
-> FROM ORDERS
-> GROUP BY SNUM
-> HAVING SUM(AMT) > 2565.84;
+-----+-----+
| SNUM | TOTAL_AMOUNT |
+-----+-----+
| 1001 | 15382.07 |
| 1002 | 5546.15 |
+-----+-----+
2 rows in set (0.009 sec)

KD4_98828_Gayatri@localhost>HAVING SUM(AMT) > 2565.84
-> ^C
KD4_98828_Gayatri@localhost>SELECT CITY, MAX(RATING) AS HIGHEST_RATING
-> FROM CUSTOMERS
-> GROUP BY CITY;
+-----+-----+
| CITY | HIGHEST_RATING |
+-----+-----+
| London | 100 |
| Rome | 200 |
| San Jose | 300 |
| Berlin | 300 |
+-----+-----+
4 rows in set (0.011 sec)

KD4_98828_Gayatri@localhost>SELECT SNUM, MAX(AMT) AS MAX_ORDER_AMOUNT
-> FROM ORDERS
-> WHERE AMT > 3000
-> GROUP BY SNUM;
+-----+-----+
| SNUM | MAX_ORDER_AMOUNT |
+-----+-----+
| 1002 | 5160.45 |
| 1001 | 9891.88 |
+-----+-----+
```

6. Retrieve the maximum order amount for each salesperson from the orders table, including only those orders whose value exceeds Rs. 3000.

```
Windows PowerShell x Windows PowerShell x Windows PowerShell x + v
--> HAVING SUM(AMT) > 2565.84;
+-----+
| SNUM | TOTAL_AMOUNT |
+-----+
| 1001 | 15382.07 |
| 1002 | 5546.15 |
+-----+
2 rows in set (0.009 sec)

KD4_98828_Gayatri@localhost>HAVING SUM(AMT) > 2565.84
--> ^C
KD4_98828_Gayatri@localhost>SELECT CITY, MAX(RATING) AS HIGHEST_RATING
--> FROM CUSTOMERS
--> GROUP BY CITY;
+-----+
| CITY | HIGHEST_RATING |
+-----+
| London | 100 |
| Rome | 200 |
| San Jose | 300 |
| Berlin | 300 |
+-----+
4 rows in set (0.011 sec)

KD4_98828_Gayatri@localhost>SELECT SNUM, MAX(AMT) AS MAX_ORDER_AMOUNT
--> FROM ORDERS
--> WHERE AMT > 3000
--> GROUP BY SNUM;
+-----+
| SNUM | MAX_ORDER_AMOUNT |
+-----+
| 1002 | 5160.45 |
| 1001 | 9891.88 |
+-----+
2 rows in set (0.014 sec)

KD4_98828_Gayatri@localhost>
```

7. Write an SQL query to find the smallest order placed by every customer

```
Windows PowerShell x Windows PowerShell x Windows PowerShell x + v
--> HAVING SUM(AMT) > 2565.84;
+-----+
| SNUM | TOTAL_AMOUNT |
+-----+
| 1001 | 15382.07 |
| 1002 | 5546.15 |
+-----+
2 rows in set (0.009 sec)

KD4_98828_Gayatri@localhost>HAVING SUM(AMT) > 2565.84
--> ^C
KD4_98828_Gayatri@localhost>SELECT CITY, MAX(RATING) AS HIGHEST_RATING
--> FROM CUSTOMERS
--> GROUP BY CITY;
+-----+
| CITY | HIGHEST_RATING |
+-----+
| London | 100 |
| Rome | 200 |
| San Jose | 300 |
| Berlin | 300 |
+-----+
4 rows in set (0.011 sec)

KD4_98828_Gayatri@localhost>SELECT SNUM, MAX(AMT) AS MAX_ORDER_AMOUNT
--> FROM ORDERS
--> WHERE AMT > 3000
--> GROUP BY SNUM;
+-----+
| SNUM | MAX_ORDER_AMOUNT |
+-----+
| 1002 | 5160.45 |
| 1001 | 9891.88 |
+-----+
2 rows in set (0.014 sec)

KD4_98828_Gayatri@localhost>
```